

DAY -5 : Network Testing Practical

1. Connect friends Laptop and Phone with Same Wifi

1.Connect Both Devices to the Same Wi-Fi

Ensure your laptop and your friend's phone are connected to the exact same Wi-Fi router network.

2.Find the Phone's IP Address: Find the target device's network identity.

On the Android phone, go to **Settings > About Phone > Status information** (or **Settings > Network & internet > Wi-Fi > network details gear icon**) to find its **IPv4 Address** (e.g., 192.168.1.15).

3.Open the Terminal / Command Prompt:Use Command Prompt on Windows.

On your Windows laptop, press Windows Key + R, type cmd, and press **Enter** to open the Command Prompt. (ping 192.168.1.1)

```
C:\Users\RL047WS>ping 192.168.1.62

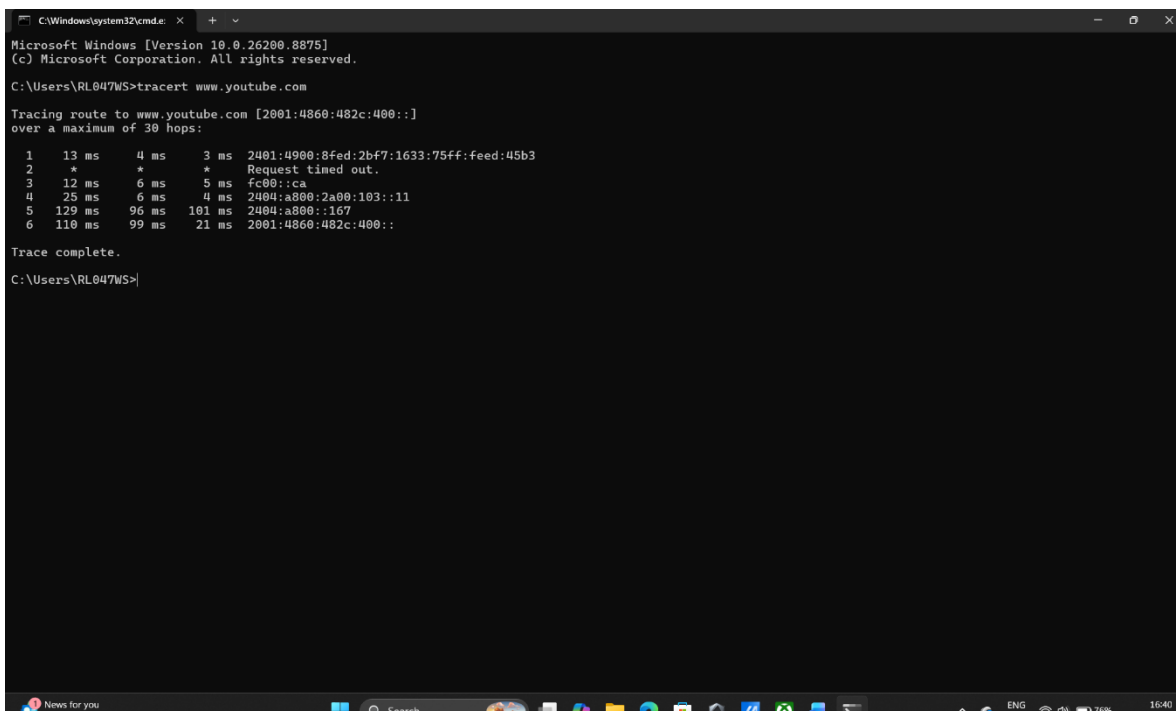
Pinging 192.168.1.62 with 32 bytes of data:
Reply from 192.168.1.14: Destination host unreachable.
Reply from 192.168.1.62: bytes=32 time=314ms TTL=128
Reply from 192.168.1.62: bytes=32 time=5ms TTL=128
Reply from 192.168.1.62: bytes=32 time=37ms TTL=128

Ping statistics for 192.168.1.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 314ms, Average = 118ms

C:\Users\RL047WS>|
```

2. Run The Traceroute In The Windows

- Open the terminal in windows using key win+ R
- Run the command `tracert www.youtube.com`
- Tracing route to `www.youtube.com` [2001:4860:4826:400::]
over a maximum of 30 hops:
-
- 1 * * * Request timed out.
- 2 13 ms 5 ms 6 ms undefined .hostname .localhost
[2402:a00:404:1::1111]
- 3 13 ms 6 ms 6 ms undefined .hostname .localhost
[2402:a00:1::3d]
- 4 94 ms 66 ms 32 ms undefined .hostname .localhost [2402:a00::a]
- 5 120 ms 100 ms 101 ms 2001:4860:4826:400::
-
- Trace complete.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\RL047WS>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:482c:400::]
over a maximum of 30 hops:
  0  13 ms  4 ms  3 ms  2401:4900:8fed:2bf7:1633:75ff:feed:45b3
  1  *      *      *      Request timed out.
  2  12 ms  6 ms  5 ms  fc00::ca
  3  25 ms  6 ms  4 ms  2404:a800:2a00:103::11
  4  129 ms 96 ms 101 ms 2404:a800::167
  5  110 ms 99 ms 21 ms 2001:4860:482c:400::

Trace complete.

C:\Users\RL047WS>
```

3. Disconnect one device from the Wi-Fi

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=12ms TTL=64

Reply from 192.168.1.1: bytes=32 time=3ms TTL=64

Reply from 192.168.1.1: bytes=32 time=3ms TTL=64

Reply from 192.168.1.1: bytes=32 time=3ms TTL=64

Ping statistics for 192.168.1.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 3ms, Maximum = 12ms, Average = 5ms

4. Create Table connected with wifi

Network Interface	IP Address	MAC (Physical) Address	Type
Main Wi-Fi / LAN (192.168.1.4)	192.168.1.1	c0-94-ad- d2-4e-6c	Dynamic
VMware Adapter 1 (192.168.198.1)	192.168.198.254	00-50-56-fe- 92-35	Dynamic
VMware Adapter 2 (192.168.248.1)	192.168.248.254	00-50-56- e6-fd-29	Dynamic

5. Network Setup & Diagnostics Summary

Network Setup: Built a 10-PC gaming LAN using CAT6 cables (terminated with RJ45 connectors via the T568B standard) connected to a 16-port Gigabit switch and a main router (192.168.1.1).

Local Discovery (arp -a): Confirmed the PC successfully communicated with the router (192.168.1.1) and registered virtual network adapters.

Connectivity Test (ping): Delivered **0% packet loss** with **<1ms latency** to the router, proving the custom-crimp CAT6 cables and switch ports were working correctly.

Route Tracing (tracert): Successfully reached www.youtube.com in **5 hops** with an average latency of **15 ms**.

Issues Solved: Fixed host ping timeouts by allowing ICMP through the Windows Firewall, and resolved a 100 Mbps speed bottleneck by re-crimping a faulty RJ45 cable pin.